

Introduction

The Honors Research project will combine fields of engineering to develop an understanding of an advanced fluidic system. The creation aims to model the physiological mechanical forces that act on a variety of natural cells. In particular, the focus is on cells within the Central Nervous System (CNS). In literature, biochemical environments of the CNS have been mapped, but mechanical environments are underrepresented in in vitro modeling. The project focuses on understanding mechanical forces, such as fluid shear stress generated by blood, cerebrospinal fluid, and interstitial flow, to develop and optimize microfluidic devices. Specifically, the project aims to improve a high-throughput 12-well cone-and-plate viscometer system. These systems serve as crucial platforms for the study of static cell culture and dynamic in vivo–mimicking environments. In the viscometer, the primary apparatus is a modified cone-and-plate device and is designed to apply a set, uniform laminar shear stress to the cultured cells. The goal of this device is to mimic the fluid shear forces at the blood-brain barrier or fluid in the ventricular systems. After fluid crosses the Blood-Brain-Barrier, the mechanical forces experienced by neuronal cells are substantially attenuated by surrounding tissue. This results in shear regimes closer to interstitial creeping flow than classical vascular hemodynamics. A distinction of this research is the specific focus on modeling very slow flows and low-magnitude shear stresses. Unlike many other works in the field that focus on high-velocity, vascular-level flows, this project targets the subtle fluid dynamics characteristic of the interstitial space and ventricular systems.

Cells in the CNS are consistently exposed to fluid flow, and standard static cultures poorly model these forces. This leads to incorrect expression of cell phenotypes and morphologies. Furthermore, existing experimental tools to study these forces suffer from reliability issues. The current 12-well cone-and-plate device in the Cornelison Laboratory relies on a suspended motor that is prone to displacement. During long experiments, studies involving shear stress can run >24 hours. The motor can become displaced due to vibrations and alter the gap height between the cone and plate. This leads to an undefined shear stress or causes stalling. Since the device is utilized within closed-door incubators, the failure can go unnoticed until the experiment is finished. This leads to failures of experiments, loss of data, and cost losses. Additionally, heat generated by the motor in a confined environment can risk the cells from overheating. One goal is the development of a securable, autoclavable motor mounting system. This system must be secured and reliably fix the motor relative to the plate. The design of this system must be able to withstand an incubator environment as well as UV sterilization.

Understanding how shear stress regulates CNS cells is important for the advancement of neurobiology and disease modeling. While the effects of high-shear environments are well-documented in cardiovascular research, the effects of creeping flow environments of the brain interstitium are often overlooked. This project is significant because it provides a reliable technological platform to study these specific, subtle forces. By enabling the precise application of low-magnitude shear stress, this research aims to mimic the hemodynamic forces at the blood-brain barrier (BBB) and the fluid dynamics of the interstitial systems. This is critical for studying pathological

progressions, such as epilepsy, where fluid dynamics may play a role in ion channel regulation, or metastasis, where fluid flow guides cancer cell migration. The successful completion of this project will contribute to the field in two distinct ways. First, from an engineering perspective, it offers an optimized solution for converting standard culture plates into sophisticated bioreactors. The introduction of pulsatile flow capabilities and live-imaging compatibility significantly expands the utility of the cone-and-plate viscometer beyond its current limitations. Second, from a biological perspective, the validated system will allow for the generation of data that decouples mechanical cues from biochemical ones. By ensuring that observed cell behaviors are the result of defined shear stress, this project will improve the reproducibility of in vitro CNS models. This reliability is essential for reducing the cost of drug discovery and improving our fundamental understanding of how physical forces dictate the health and disease states of the central nervous system.

Summary of work of previous researchers

Objective

This literature review aims to understand engineered mechanofluidic systems to model the physiological fluid flow and shear stress experienced by cells in the

Central Nervous System (CNS). Vascular hemodynamics are well-characterized across various systems; modeling very slow flows and low-magnitude shear stresses presents unique engineering challenges. The review analyzes the evolution of high-throughput cone-and-plate viscometers, peristaltic perfusion loops, and transwell microfluidic devices. The review will identify design optimizations that improve the reliability of these systems for modeling the Blood Brain Barrier (BBB), ependymal cilia mechanics, and brain metastasis.

Methods

This literature review compares eleven peer-reviewed sources. Of those sources, ten are original research articles analyzing in vitro device engineering, and one is an in vivo study utilizing artificial intelligence velocimetry. Four sources focus on cone-and-plate mechanics, three on peristaltic loop systems, and four on transwell-integrated designs. The main question of "How can engineered systems accurately model low-magnitude CNS shear stress" was tracked. Sources [1], [2], [3], and [4] were utilized to determine the geometric and feedback mechanisms required for uniform shear application. Subsequently, sources [6], [7], and [11] were reviewed to define the boundaries of creeping flow and establish laminar dampening protocols. Sources [8], [9], and [10] were analyzed to explore design strategies for integrating barrier integrity assays (TEER) into dynamic environments. Finally, source [5] was used to contextually link mechanical forces to pathological cell phenotypes. Listed sources are

referenced multiple times throughout the review, and are not limited to their specific subsection.

Sources

The eleven sources chosen for this literature review came from several journal sources. Sources [7] and [9] come from the journal *Lab on a Chip*. Sources [1], [2], and [3] come from *Computers in Biology and Medicine, Biotechnology and Bioengineering, and Cytotechnology*, respectively. Source [6] is from the *Journal of Biological Chemistry*. Source [5] is from *Scientific Reports*, while Source [4] is from the *European Journal of Cell Biology*. Sources [8] and [10] are from *ACS Measurement Science Au* and *Frontiers in Bioengineering and Biotechnology*, respectively. Finally, Source [11] is from the *Proceedings of the National Academy of Sciences*. All were listed as scholarly journals, and each source was reviewed for information regarding mechanofluidic engineering and CNS modeling.

Conclusion

The review highlights a rapid maturation of mechanofluidic technologies for modeling CNS fluid dynamics. Cone-and-plate systems have achieved exceptional control over shear gradients, temporal waveforms, and active feedback, enabling detailed studies of endothelial mechanotransduction [1], [2], [3], [4]. Closed-loop

peristaltic systems provide reproducible suspension culture and circulatory stress models, revealing how shear stress can drive hybrid epithelial, mesenchymal, and stem-like phenotypes [5]. [6], [7]. Transwell-integrated millifluidics balance barrier integrity with dynamic flow, illustrating the importance of multicellular interactions in the blood-brain barrier [8,9]. Hybrid devices like the MoCAP extend these capabilities, combining high-throughput shear application with TEER and permeability readouts [10]. However, in vivo measurements by Boster et al. reveal that perivascular shear stresses in the CNS are orders of magnitude lower than most in vitro systems currently applied [11]. This critical insight underscores the need for next-generation devices capable of generating ultra-low, creeping flows to accurately replicate CNS interstitial and perivascular environments. Future research should focus on refining device geometries, flow control mechanisms, and computational validation to produce truly physiologically relevant models, ultimately enhancing our understanding of CNS hemodynamics, barrier function, and disease progression.

Literature Review Introduction

The fluidic environment of the Central Nervous System (CNS) is a complex system of forces distinct from the body's vascular hemodynamics. While high-velocity, pulsatile flow of blood into the neurovasculature is critical for oxygen transport, there exists extremely low magnitude, diffusion-dominant flow of cerebrospinal fluid (CSF) in the system. These forces play a vital role in waste clearance and metabolic homeostasis.

Disruptions in these fluidic profiles are linked with increased risk of neurodegenerative diseases, metastatic colonization, and blood-brain-barrier (BBB) dysfunctions.

Consequently, the ability to model these specific hemodynamic and hydrodynamic forces in vitro is a prerequisite for developing an accurate model for CNS diseases.

Historically, the testing of physiological shear stress relies on three engineering approaches: the cone-and-plate viscometer, peristaltic perfusion loops, and microfluidic transwell systems. The cone-and-plate viscometer system is the longest-standing device. This device applies a uniform shear stress to cell monolayers, with early iterations focused on establishing controlled hemodynamic distributions [1]. Newer iterations have the power of high-throughput capabilities to study endothelial responses [2], and current systems can reproduce realistic shear stress waveforms, used to understand cellular behaviors and molecular adaptation to circulatory forces. [3], [4], [5]. However, traditional cone-and-plate designs often struggle to integrate porous substrates necessary for modeling tissue barriers, which limits their ability in BBB research.

The peristaltic pump-based system is another system to model shear stresses. This device introduces a laminar flow into tubular and channel devices and is tunable for media circulations [6], [7]. These devices have been used to study the survival and selection of circulating tumor cells under shear [5]. However, these systems inherently require more complex dampening systems to mimic smooth, low-velocity flows found in the CNS [6]. Recent developments in the field have pushed more towards transwell-based millifluidic devices and 3D-printed systems [8], [9]. These newer devices try to preserve the ability of transwells to form robust cellular barriers, while

also adding fluid flow to better replicate mechanical forces present in living tissues. While these innovations improve physiological relevance, they often lack the spatial and shear-stress resolution required to model the creeping-flow regimes characteristic of interstitial and perivascular transport in the CNS.

A critical gap remains in the engineering of systems that can simultaneously support barrier integrity and replicate the ultra-low shear stresses specific to the CNS. This review analyzes the evolution of these mechanofluidic technologies, with a specific focus on recent hybrid designs. Notably, new modular cone-and-plate lids have been developed to retrofit static transwell inserts with dynamic shear capabilities [10]. Furthermore, the validation of these systems has been revolutionized by artificial intelligence velocimetry, which recently has provided the first in vivo quantification of the minute shear stresses present in perivascular flows [11]. This paper will evaluate these emerging design optimizations and their application to modeling BBB integrity, ependymal cilia mechanics, and brain metastasis.

To contextualize the engineering constraints of these systems, it is first necessary to define the physical basis of fluid shear stress. Fluidic shear stress (τ) is the tangential force per unit area exerted by a fluid moving across a solid boundary. In the context of vasculature, the force is applied by blood flow to the endothelial glycocalyx (outer surface). The magnitude of the stress is determined by the viscosity of the blood (μ) and the shear rate ($\dot{\gamma}$), which is the gradient of velocity perpendicular to the wall. To

replicate this force in vitro (outside a living organism), there must be precision controls over the fluid velocity profiles.

Evolution of Cone-and-Plate Technologies

The cone-and-plate apparatus has evolved from a simple rheological tool into a sophisticated platform for high-throughput mechanobiology. Ye et al. addressed the geometric limitations of standard cones by designing a truncated generatrix cone. Through Finite Element Method simulations, they demonstrated that standard cones often produce variable shear forces near the center of the well due to singular points of contact. By modifying the cone geometry to include a truncated tip and optimizing the gap height to 0.1 mm, Ye et al. achieved a shear stress field where over 69% of the culture area experienced a uniform force of 1–2 Pa. This specific optimization ensures that the majority of the cell monolayer is subjected to identical mechanical cues, reducing the noise in biological readouts that often plagues standard plate-based assays. They demonstrated that by optimizing the gap height and cone geometry, these devices can establish highly controlled distributions of wall fluid shear stress [1].

While Ye et al. focused on single-well precision, Spruell and Baker developed a high-throughput 96-cone apparatus. The primary engineering challenge in this technology is maintaining a consistent gap height (h) across a large plate surface, as minor deviations can drastically alter the applied shear. To solve this, Spruell and Baker integrated a micropositioning stage capable of micrometer-level adjustments to the

plate's vertical position relative to the cone array. They characterized the device's performance at varying gap heights, revealing that a 100 μm gap produces a highly linear radial gradient, whereas a 1 μm gap approximates an ideal uniform shear profile. This capability allows researchers to utilize a single 96-well plate to screen both uniform conditions and gradient thresholds simultaneously [2].

Recent advancements have moved beyond steady-state flow to replicate the complex dynamics of the cardiac cycle. Franzoni et al. developed a compact, programmable device capable of generating realistic, unsteady shear stress waveforms inside a standard incubator. Validating the device against analytical solutions, they defined a radius of uniformity greater than 25 mm where shear stress is directly proportional to angular velocity. Crucially, they demonstrated the ability to reproduce oscillating flow conditions at frequencies up to 10 Hz. This allows for the modeling of disturbed flow, a hemodynamic state characterized by low mean shear and oscillatory direction changes. This scenario is known as a driver of atherosclerosis and aneurysm formation in the CNS vasculature [3]. The integration of real-time feedback represents the latest leap in this technology.

Huang et al. introduced a torque sensor-coupled mechanism that actively monitors the resistance encountered by the cone. This feedback loop allows for the precise maintenance of stable laminar shear stress up to 50 dyne/cm^2 , correcting for viscosity changes or mechanical drift over long-term culture. The biological validation of this system using urothelial carcinoma cells provided a mechanistic link between mechanics and cell fate: stable shear at 12 dyne/cm^2 induced G1 cell cycle arrest via the

upregulation of p21 and p27 (CDK inhibitors) and triggered autophagy markers Beclin-1 and LC3B [4]. By comparing these four systems, we see a clear trajectory from geometric to throughput, temporal complexity, and finally active feedback control, providing a robust toolkit for adherent cell mechanobiology [1], [2], [3], [4].

Peristaltic and Closed-Loop Systems

In contrast to cone-and-plate systems, which are open and apply shear to adherent cells, peristaltic pump-based loops are closed systems essential for modeling circulating transport and long-term suspension culture. However, a major engineering limitation of peristaltic pumps is the generation of pulsatile artifacts, where rhythmic pressure spikes are caused by the rollers compressing the tubing, which can confound studies of laminar flow. Abello et al. conducted rigorous comparative analysis of dampening mechanisms to try to solve this issue. They engineered custom inlet and outlet dampeners and compared them against commercially available options. The results were striking: while commercial dampeners achieved only a 1.4-fold suppression of fluid pulsation, the custom-built dampeners achieved a six-fold suppression. Particle tracking velocimetry confirmed that the custom dampeners virtually eliminated the rhythmic shaking of suspended particles, allowing standard laboratory pumps to generate true laminar flow. This design of laminar flow capability is critical for labs that cannot afford specialized syringe pumps [6]. Taking a different approach to flow control, Schneider et al. developed the Organ-Disc, a microfluidic platform that integrates the pumping mechanism directly onto the chip. Instead of external rollers, the Organ-Disc utilizes a periodic compression of a flexible pump layer embedded within the disc itself.

This design eliminates the need for external tubing, significantly reducing the system's dead volume, which is the volume of media in the tubes that never reaches the cells. This is a crucial advantage when working with expensive reagents or limited patient-derived samples. Furthermore, the system was validated to support the perfusion of whole blood without hemolysis, a feat often difficult in external pumping systems due to the high shear forces in narrow tubing connectors. The biological necessity of these closed-loop systems is best exemplified by their application to metastasis research.

Alvarado-Estrada et al. conducted a study using a closed-loop fluidic system to expose lung adenocarcinoma cells to 72 hours of circulatory shear stress. Unlike traditional adherent models, this suspension system subjected the cells to continuous mechanical forces, creating a strong selective pressure. Interestingly, the surviving cells did more than resist the stress—they adapted. They exhibited a hybrid epithelial–mesenchymal phenotype, characterized by increased expression of the transcription factors TWIST1 and SNAI2, which promote migration, while retaining the epithelial adhesion marker EpCAM. Notably, shear stress selectively enriched a stem-like side population of cells, increasing their proportion by fourfold compared to static controls. This enrichment was associated with upregulation of CD44, whereas other stem markers, such as CD133 and SOX2, remained unchanged or decreased. These results illustrate that closed-loop fluidic systems function not only as transport models but also as experimental platforms that can replicate the phenotypic evolution of metastasizing cells in circulation [5].

Transwell-Integrated Millifluidics

While cone-and-plate and peristaltic systems excel at applying shear, they often struggle to model the multicellular architecture of the Blood-Brain Barrier (BBB), which requires compartmentalized co-culture of endothelial cells, pericytes, and astrocytes. Harding et al. addressed this by developing a millifluidic device that is a retrofit with standard 24-well Transwell® inserts. The design allows researchers to utilize pre-validated commercial inserts while introducing dynamic flow. Using a pulsed damper similar to the concept proposed by Abello, Harding et al. applied a laminar shear stress of 12 dyne/cm² to a human brain microvascular endothelial cell monolayer. The results highlighted the effects of mechanotransduction in multicellular environments: while shear stress significantly reduced dextran permeability and tightened the barrier in endothelial mono-cultures, it had negligible effects on barrier integrity in co-cultures containing pericytes and astrocytes. This suggests that the presence of supporting cells may mechanically or chemically buffer the endothelium, a finding that would be impossible to observe in a standard cone-and-plate device lacking a basolateral compartment [9]. This finding suggests that cellular interactions may stabilize barrier function against mechanical changes.

Pushing the engineering further, Cenhrang et al. utilized vat polymerization 3D printing to create a fully custom transwell bioreactor. Unlike Harding's retrofit, this device is built from the ground up to optimize imaging and sealing. It features a unique screw-top reservoir that is open during cell seeding but sealed with an M4 screw during flow application. Thus, it can enable a transition from static to dynamic culture without

moving the cells. A key material innovation was the use of an electrospun collagen scaffold sandwiched between laser-cut Teflon membranes. These provide a more physiological substrate than standard polycarbonate filters. The device was validated with flow rates ranging from 0.015 to 0.5 mL/min and demonstrated compatibility with Potentiometric Scanning Ion Conductance Microscopy, allowing for high-resolution topographical mapping of the barrier under flow [8]. This contrasts with the Harding device, which prioritizes ease of use and throughput, whereas the Cenhrang device prioritizes substrate fidelity and imaging resolution

Hybrid Designs and AI Validation

The pairing of these technologies is seen in the Modular Cone-and-Plate (MoCAP) device developed by Chavarria et al. This system represents a hybrid approach, applying the shear precision of a cone-and-plate viscometer onto a standard Transwell® insert. Fabricated using high-temperature resin via SLA 3D printing, the MoCAP features a modular gearbox driven by a stepper motor. The device is relatively cheap and easy to manufacture, but the ingenuity lies in the interchangeability of the design. By swapping 3D-printed gears, the device can generate a spectrum of shear stresses (0.6 to 20 dyn/cm²) across different columns of the same plate simultaneously. Crucially, the cones are coated in Parylene-C to ensure biocompatibility and are positioned with a precise 200 µm gap above the transwell membrane. This design solves the barrier gap of traditional viscometers described by Ye and Spruell. His configuration

allows for TEER (transendothelial electrical resistance) measurements, which assess the integrity of the cell barrier by measuring electrical resistance across the cell layer, and permeability assays, which measure the passage of molecules through the barrier to determine how leaky or intact it is. Together, these assays provide a comprehensive evaluation of barrier integrity under dynamic shear conditions while the device maintains the high-throughput shear gradients typical of cone-and-plate systems. Despite these advantages, the effectiveness of any in vitro system ultimately depends on how well it replicates in vivo conditions, which is a significant challenge in modeling the central nervous system. Boster et al. provided a critical benchmark using Artificial Intelligence Velocimetry (AIV). By training Physics-Informed Neural Networks on sparse particle tracking data, they were able to reconstruct the 3D flow fields of the murine with unprecedented resolution. The findings exposed that the mean flow speed in the perivascular space is approximately $16.33 \mu\text{m/s}$, which results in wall shear stresses between 0.015 and 0.050 dyn/cm^2 . This is orders of magnitude lower than the $10\text{--}20 \text{ dyn/cm}^2$ typically applied in vascular models. This discrepancy highlights a critical physiological mismatch in current Glymphatic-on-Chip models. It suggests that while systems like the MoCAP are capable of generating low shear (down to 0.6 dyn/cm^2), even these lower limits may be too high for accurately modeling the interstitial spaces of the brain [11]. The data from Boster et al. serves as the ultimate calibration standard, driving the next generation of device engineering toward ultra-low, creeping flow regimes.

Conclusion

The studies reviewed collectively demonstrate a rapid maturation of mechanofluidic technologies, moving from simple fluid agitation to precision mechanobiology. The cone-and-plate lineage [1], [2], [3], [4] has achieved exceptional control over shear gradients and waveforms, making them ideal for dissecting endothelial signaling pathways. Meanwhile, closed-loop peristaltic systems [5], [6], [7] have become indispensable for modeling the fluid profile of metastasis, revealing how circulation selects for hybrid EMT/stem-like phenotypes. In the realm of barrier modeling, the field is splitting between accessible retrofits [9] and highly specialized custom fabrications [8], with hybrid designs like the MoCAP [10] attempting to offer the best of both worlds. However, the AI simulations provided by Boster et al. reveal that our engineered systems are often overpowering the cells with non-physiological forces when modeling the CNS specifically [11]. The discovery that perivascular shear is as low as 0.015 dyn/cm^2 challenges the field to develop new pumping and dampening strategies capable of stable ultra-low flow. Future developments must focus on bridging this final gap by downsizing the precision of MoCAP and Organ-Disc to the microscopic regime defined by AI velocimetry, thereby creating truly predictive models of CNS hemodynamics and disease.

Methods

Legacy Device

The legacy 12-well cone-and-plate viscometer comprised six components: a base plate, 12-well cell culture plate, cone holder, 12 cones, 12 gears, and a motor mount. Three problems motivated the redesign. First, the motor mount used a cantilevered frame secured with an elastic band, leaving the motor unsecured during operation and making setup technically demanding for laboratory technicians. Second, metal shavings were consistently recovered from the culture media following overnight experimental runs, indicating friction-induced particulate contamination within the drive assembly. Third, the existing method for setting the cone gap height lacked sufficient precision for reliable shear stress application. An additional design goal was to enable live-cell fluorescence imaging from beneath the base plate, which placed constraints on the footprint and geometry of any drive mechanism.

Design Approach

The redesign followed an iterative engineering cycle in which each prototype was subjected to long-duration experimental trials before refinements were identified and implemented. All components were modeled in Onshape and prototyped via fused deposition modeling (FDM) using polylactic acid (PLA). The progression of design decisions, failure modes encountered, and the rationale for each subsequent iteration are reported in the Discussion.

Standard operation of the legacy device began by loading the 12-well plate with cells and culture media beneath the plate stand (Figure 2). The 12 cone-and-gear assemblies were then installed in a deliberate snake-like pattern dictated by the gear

geometry. Because the gears had no positional tolerance and used an even tooth count, each cone had to be placed precisely between its neighbors, and any misalignment caused the lower lips of adjacent cones to contact one another, preventing rotation (Figure 1). Critically, this insertion process required significant care from the technician, as any agitation of the plate risked disturbing the cell layer settled at the bottom of each well; excessive contact pressure was sufficient to mechanically destroy the cells before the experiment began.



Figure 1. Cone and Gear Mesh

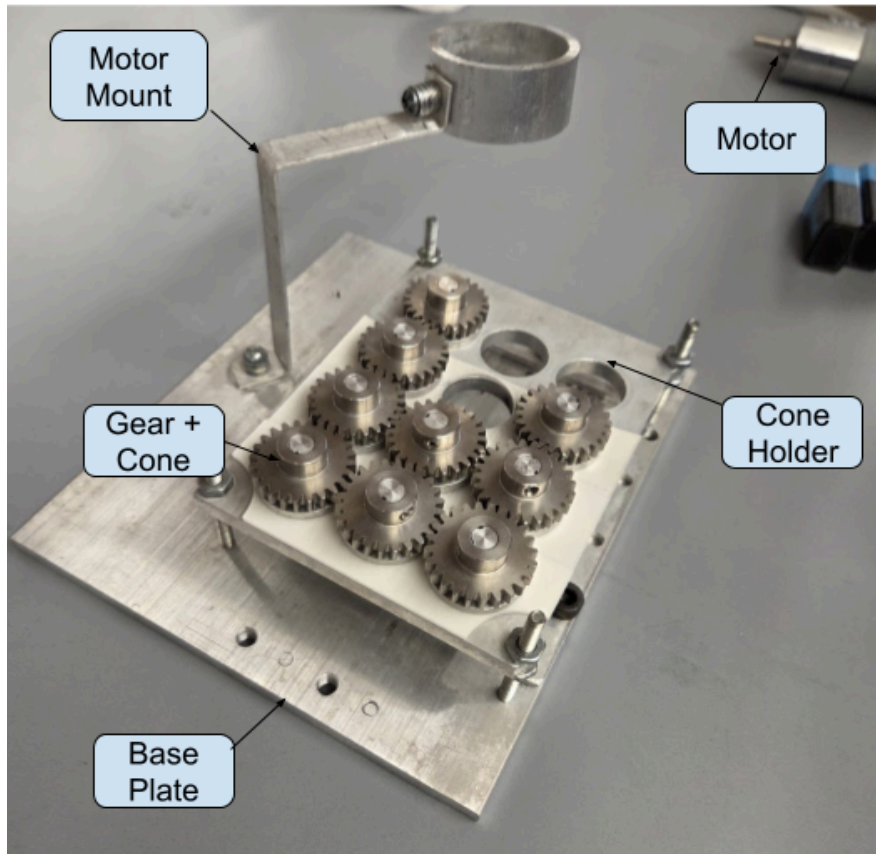


Figure 2. Existing 12-well cone setup

Physical examination of the device revealed two compounding issues. First, the cone and gear assemblies had been press-fitted together as a single aluminum unit with no designed separation mechanism. There was an idea to disassemble them using thermal cycling. But they were deemed unsuccessful, as both components shared the same material and therefore the same expansion coefficient. Mechanical separation was

not pursued since there was a risk of damaging the cone. Second, the drive train consisted of 12 gears in series, powered by a single input gear. Each gear consisted of an even tooth count, which increased the probability that the same tooth faces re-engage repeatedly under load. This leads to concentrated wear rather than distributing it across the full gear face. This makes jamming more likely than with an odd tooth count. With 12 gears driven in series, this wear was compounded by the cumulative torque demand across the train.

Results and Discussion

The first proposed solution aimed to resolve the motor mount instability while simultaneously enabling vertical optical scanning of the cells from below. In the legacy device, the cantilevered mount left the motor unsecured during operation, requiring the technician to wedge an elastic band between the motor shaft and the retaining cylinder to hold it in place. To address this, a side-entry drive assembly was designed utilizing a bevel gear pair to transmit torque laterally from a side-mounted motor rather than from above (Figure 4, Appendix A). During prototyping and testing, however, the bevel gear failed to deliver adequate torque to drive the full 12-cone assembly. When tested at rotational speeds below the required operating rate, the gear skipped under load and jammed intermittently, and the approach was abandoned.

A multi-functional motor mount compatible with 80-20 aluminum extrusion framing was subsequently developed to accommodate either a top-down or side-entry

drive configuration (Figure 3). Initial testing of the top-down cap configuration showed that the motor and cap could successfully initiate rotation of the 12-gear system. However, after sustained operation at 1.5 RPM for one hour, the set screw securing the cap progressively loosened, causing the cap to slip on the shaft and the motor to lose drive engagement.

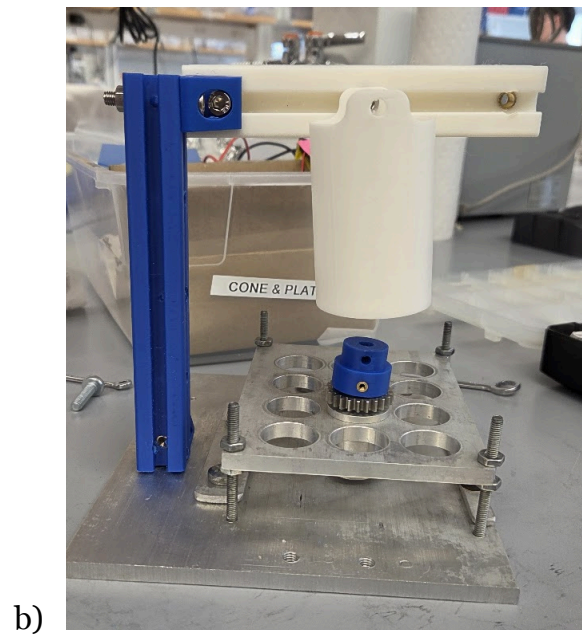
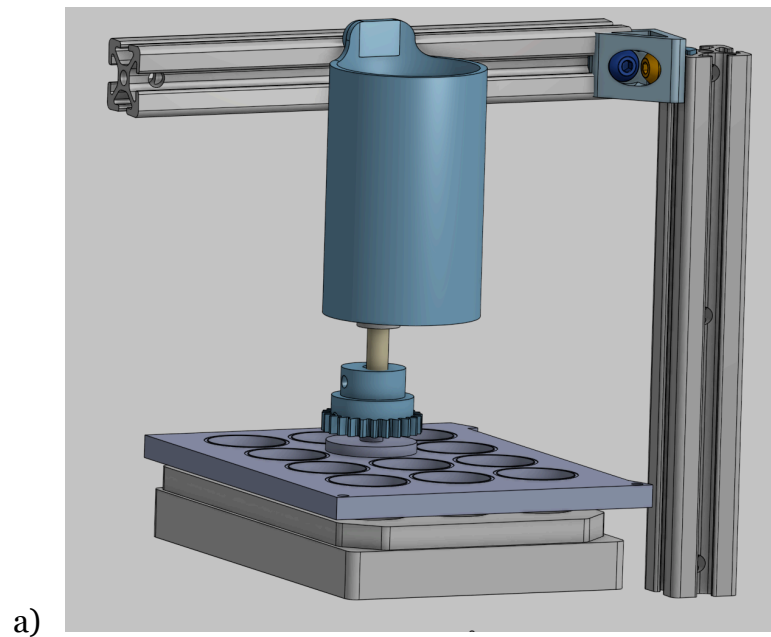


Figure 3. Top-down motor mount assembly with 80-20 extrusion holder. (A) CAD model. (B) Physical assembly.

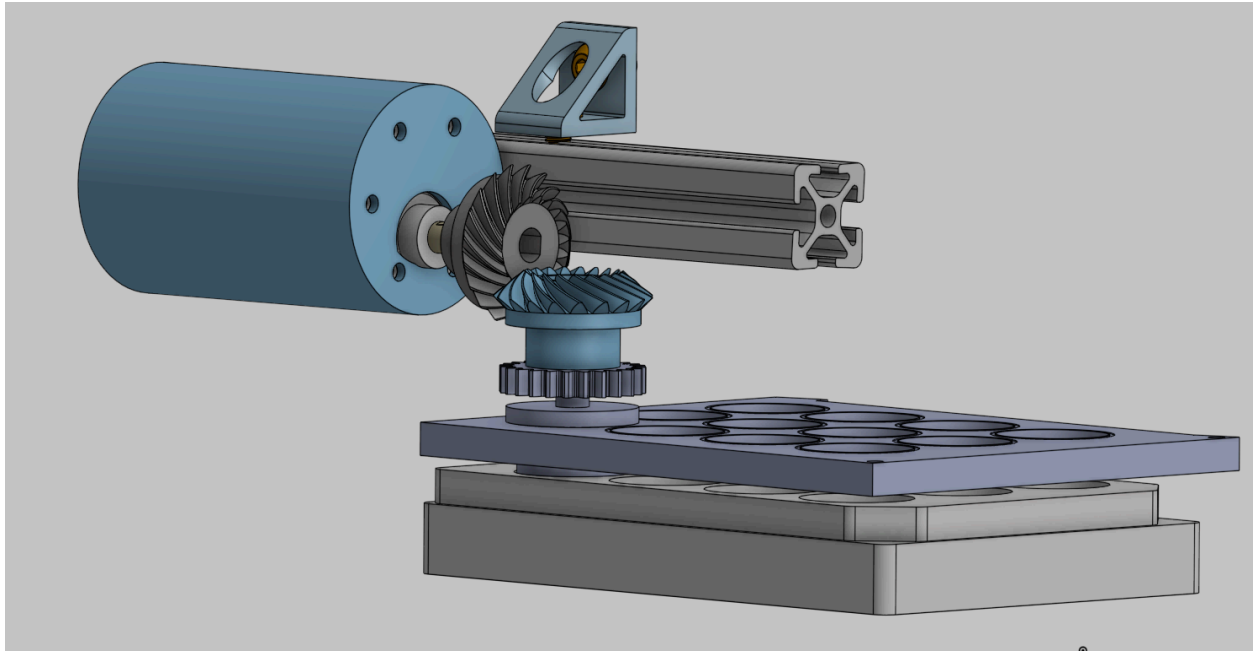


Figure 4. Side-entry bevel gear drive configuration

The following iteration employed a dual 80-20 extrusion frame in which the full assembly was secured under load, with the motor mounted vertically on the upper frame member and driving the fixed motor interface point (Figure 5). Under dynamic loading, the frame exhibited lateral wobble that introduced instability into the drive train. The number of components required also substantially increased setup time and complexity, making the configuration unsuitable for routine laboratory use. These compounding failure modes indicated that an adjustable design was insufficient and that a ground-up redesign of both the drive mechanism and the mounting system was necessary.

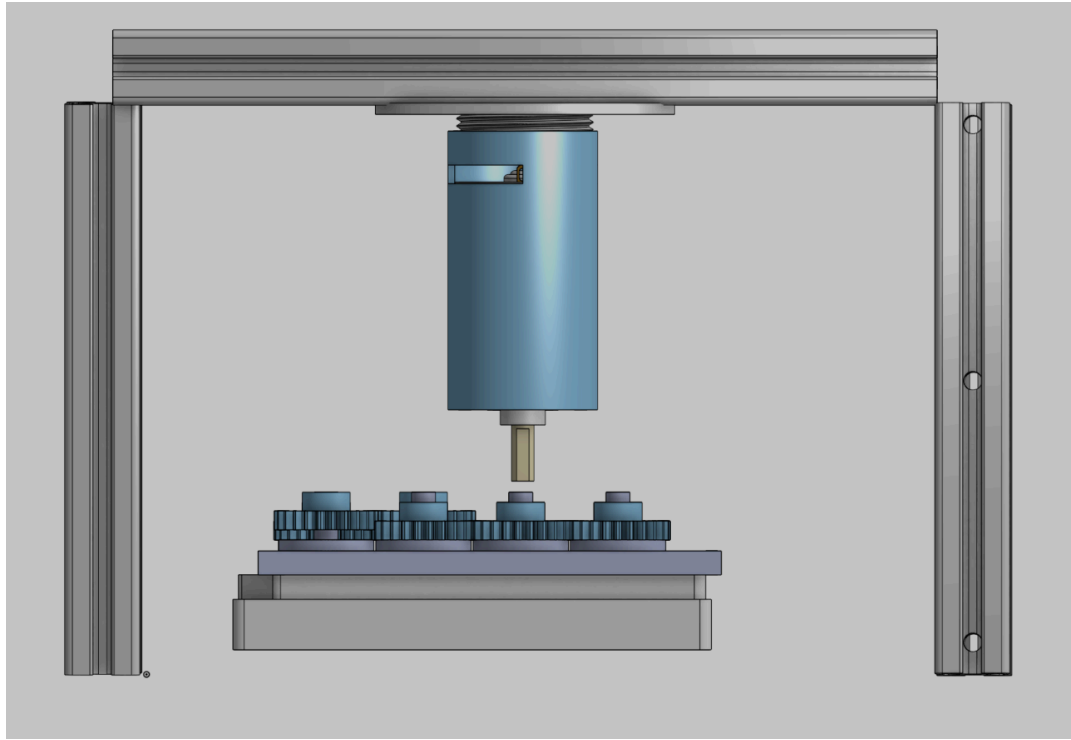


Figure 5. Dual 80-20 vertical setup

The subsequent iteration addressed the structural instability identified in prior designs by replacing the 80-20 extrusion frame with a rigid PLA portal frame (Figure 6). The motor was enclosed in a cylindrical PLA housing secured within the frame, eliminating lateral wobble and fixing the drive point to a single stable location. Under dynamic loading at the required operating speed, the assembly performed without the vibration or drive disengagement observed in previous iterations. However, following overnight experimental runs, metal shavings were again recovered from the culture media.

The source of particulate contamination required systematic investigation. Two hypotheses were considered: shearing at the gear-mesh interfaces, or shearing at the

junction between the cone and its upper retaining plate. Because frame wobble had been eliminated, vibration-induced wear was ruled out as a contributing factor. Examination of the cone holder identified the upper retaining plate as the primary source of debris, as friction between the rotating cone and the fixed plate was generating metal particulate during extended operation. Replacement of the retaining plate and redesign of the cone holder interface were established as priorities for the subsequent iteration.

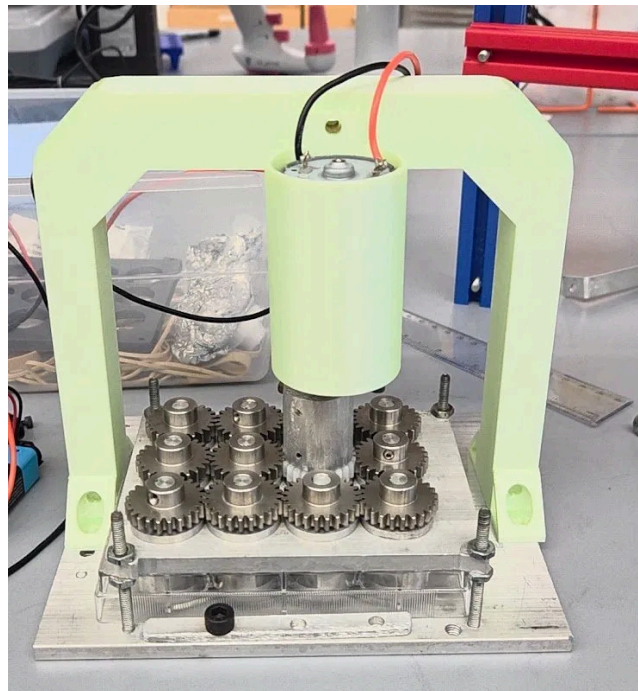
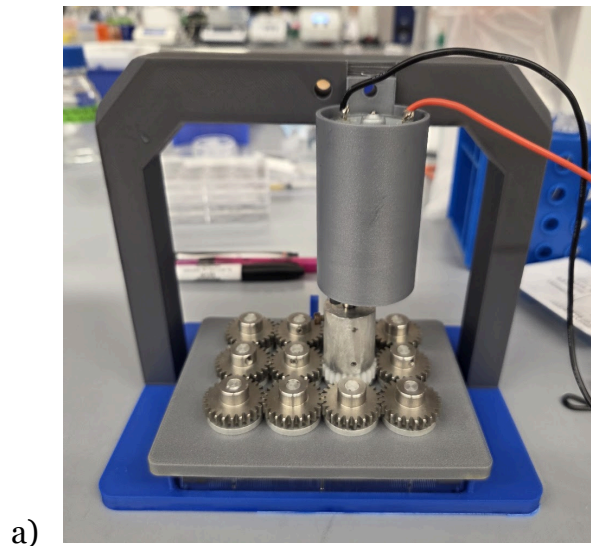
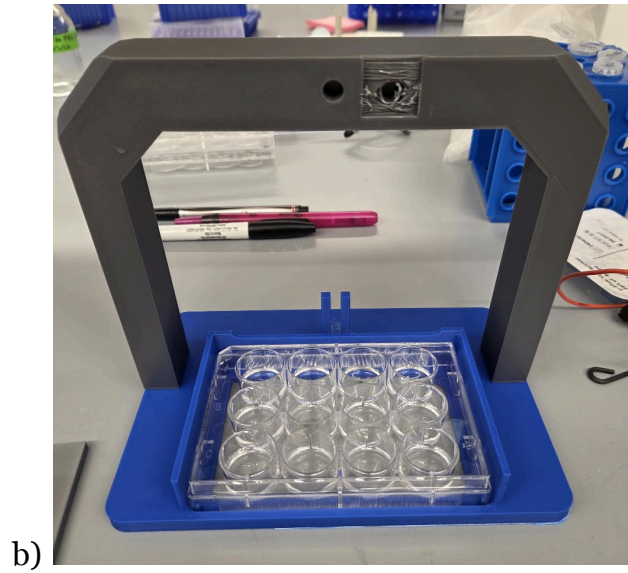


Figure 6. Rigid PLA frame motor mount

Separately, inspection of the legacy base plate revealed that the height adjustment mechanism, which relied on precision screws and nuts to set the cone gap, had failed. The nuts were stripped and could not be removed, and the base plate and cone holder were found to be unlevel, meaning the cone gap could not be reliably

controlled across all 12 wells. The subsequent redesign replaced the legacy base plate and cone holder with a PLA-printed assembly featuring a fixed-height geometry, designed to passively maintain the required 100 μm cone gap without manual adjustment (Figure 7). This configuration allowed technicians to assemble the cones independently of the well plate, significantly reducing setup time, minimizing agitation of the cell layer during installation, and eliminating the variability introduced by the legacy manual adjustment procedure.





*Figure 7. Top-down motor mount assembly with fixed motor holder. (A) Full assembly.
(B) Base plate assembly without cone holder*

Following overnight experimental runs with the redesigned assembly, post-trial inspection revealed that the cone gap height was inconsistent across wells. Investigation of the GenClone multiwell plate specification sheet revealed a manufacturing tolerance of $\pm 150 \mu\text{m}$ in the well geometry [12]. Given that the target cone gap height is $100 \mu\text{m}$, a tolerance of this magnitude means the actual gap could vary by more than the gap itself across wells within a single plate. To quantify the impact of this variability, the cone-and-plate shear stress relationship was used to plot the change in shear stress $\Delta\tau$ as a function of actual gap height h_0 at the operating speed of 2.4 RPM (Figure 8). The resulting curve demonstrated that deviations from the $100 \mu\text{m}$ target produce significant shear stress differentials across the physiologically relevant range, with errors approaching $\pm 0.05 \text{ Pa}$. Full gear ratio calculations for slow speed shear regimes

are provided in Appendix C. This magnitude is sufficient to meaningfully alter the hemodynamic stimulus experienced by the cell monolayer. This finding established that reliable experimental outcomes required gap height control well within the plate's native manufacturing tolerance. In response, critical components were redesigned for CNC machining to achieve the necessary dimensional precision. Analytical characterization of well-to-well geometric variation across multiple 12-well plate batches remains ongoing, and experimental validation of the gap-setting procedure is identified as a key next step before full biological trials can be conducted.

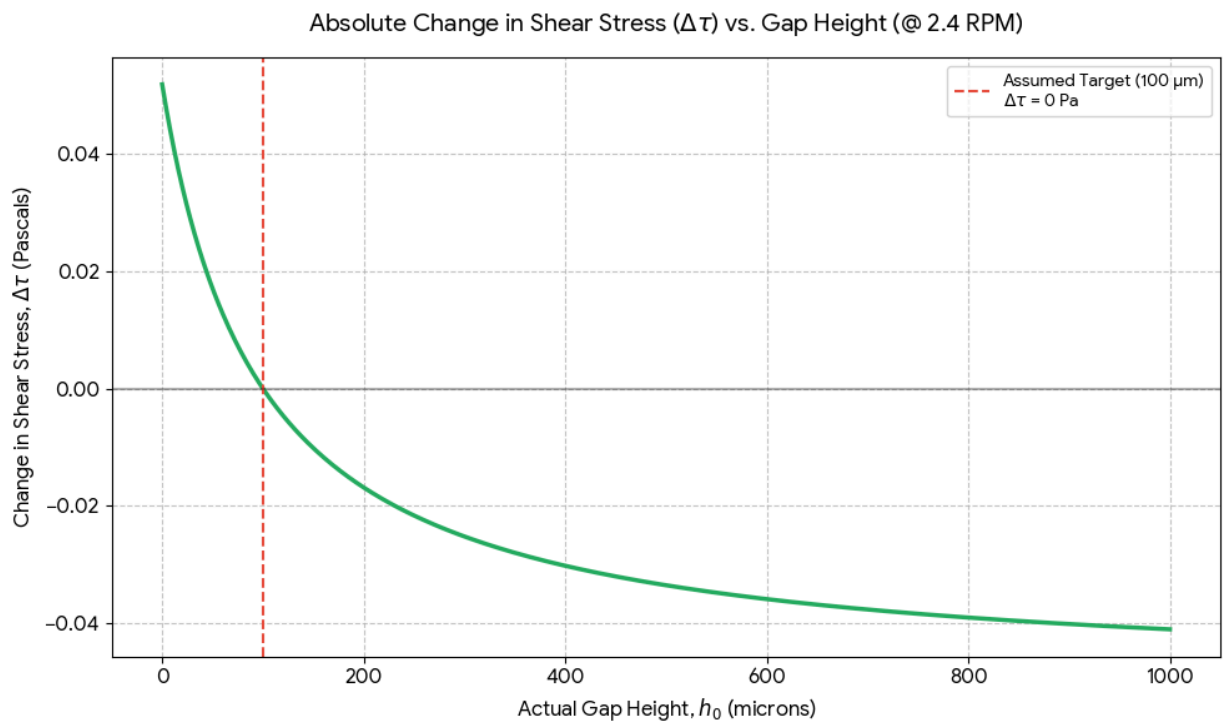


Figure 8. Change in Shear Stress relative to Gap Heights with a motor at 2.4 RPM, $a = 0.5 \text{ deg}$, $\mu = 0.00354 \text{ Pa}\cdot\text{s}$

Conclusions and Future Work

This project addressed three compounding failure modes in the legacy 12-well cone-and-plate viscometer: motor mount instability, friction-induced particulate contamination of cultures, and insufficient cone gap-setting precision. Through iterative prototyping, the cantilevered motor mount was replaced with a rigid PLA portal frame that eliminated macro-scale vibrations and drive disengagement during overnight operation. The source of metal shavings was identified as the CNC-machined upper retaining plate of the cone holder, and its replacement was established as a design priority. The base plate and cone holder were redesigned as an integrated PLA assembly, reducing setup complexity and minimizing technician-induced cell agitation during installation.

The most significant finding of this project was the discovery that gap height deviations as small as 100 μm produce shear stress errors sufficient to compromise the physiological validity of CNS hemodynamic modeling. This reframed the core engineering challenge from mechanical stabilization to sub-100 μm dimensional precision. This is a tolerance regime that exposed fundamental limitations in the current PLA-based fabrication approach.

Future Work

The immediate next step is a systematic analytical characterization of the GenClone 12-well plate geometry. Well-to-well variation within a single plate and batch-to-batch variation across multiple plates will be measured to determine whether

gap height error is consistent and predictable or random. This distinction is critical: predictable variation can be corrected by design, while random variation requires per-plate calibration.

Concurrently, the dimensional accuracy of the current PLA-printed components will be characterized. FDM printing in PLA carries an inherent tolerance of approximately ± 0.2 mm per printed feature, an error twice the magnitude of the target 100 μm cone gap. This marks it as unsuitable as a long-term material for gap-critical components without post-print measurement and correction. Additionally, PLA parts printed at less than 100% infill exhibit reduced stiffness and are susceptible to creep under the sustained mechanical loads present during overnight operation, which may introduce further dimensional drift over time.

To achieve the required sub-100 μm precision, future iterations will transition critical components to CNC-machined designs. The target design is intentionally simple and compact: a single precision spacer inserted into one well of the plate, replacing the motor-driving cone, which supports the cone holder at a fixed, precisely measured height. With the holder elevation established by this reference spacer, the remaining 11 cones are passively positioned at the correct 100 μm gap above their respective well floors. If well-to-well geometric variation is found to be within acceptable limits, this approach eliminates the need for per-well gap adjustment and provides a mechanically deterministic solution to the gap-setting problem. Design concepts for this spacer-based system have been developed and are currently under review for fabrication.

Objectives

Once dimensional precision and mechanical reliability are validated, the device will be used to conduct its first controlled biological trial. CNS endothelial cells will be subjected to physiologically relevant low-magnitude shear stress profiles, with cell morphology, gene expression, and barrier integrity measured as primary outcomes. This experiment will serve as the functional validation of the full system and will generate the first mechanobiological data from the redesigned platform.

Additionally, the redesigned base plate was specifically engineered to permit live-cell fluorescence imaging from beneath the culture wells during shear application. Compatibility of the assembled device with inverted microscopy will be formally validated, enabling real-time observation of cell response to shear without interrupting experimental runs. Together, these biological milestones represent the transition of this project from an engineering redesign effort to a fully operational research platform for CNS hemodynamic modeling.

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Annotated References

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This study utilizes Finite Element Method simulations to optimize the geometry of the cone in a cone-and-plate device. The authors introduce a truncated generatrix design, which significantly improves the uniformity of the shear stress field, achieving a uniform 1-2 Pa force over 69% of the culture area, addressing the issue of heterogeneous forces in standard designs.

[2] C. Spruell and A. B. Baker, "Analysis of a High-Throughput Cone-and-Plate Apparatus for the Application of Defined Spatiotemporal Flow to Cultured Cells," *Biotechnol. Bioeng.*, vol. 110, no. 6, pp. 1782–1793, June 2013, doi: 10.1002/bit.24823.

This paper presents the engineering of a 96-well cone-and-plate apparatus. The key innovation is the integration of a micropositioning stage that controls the gap height with high precision. The study characterizes the relationship between gap height and shear profile, demonstrating that a 100 μm gap creates a linear gradient while a 1 μm gap creates a uniform field, enabling high-throughput combinatorial screening.

[3] M. Franzoni et al., “Design of a cone-and-plate device for controlled realistic shear stress stimulation on endothelial cell monolayers,” Cytotechnology, vol. 68, no. 5, pp. 1885–1896, Oct. 2016, doi: 10.1007/s10616-015-9941-2.

Franzoni et al. focus on the temporal aspects of shear stress, developing a programmable device capable of generating disturbed flow waveforms (oscillating up to 10 Hz) within a standard incubator. The study defines the radius of uniformity (>25mm) and validates the system's ability to model atheroprone hemodynamic conditions over 48 hours.

[4] S.-Y. Huang et al., “Stable laminar shear stress induces G1 cell cycle arrest and autophagy in urothelial carcinoma by a torque sensor-coupled cone-and-plate device,” Eur. J. Cell Biol., vol. 103, no. 4, p. 151451, Dec. 2024, doi: 10.1016/j.ejcb.2024.151451.

This study integrates a torque sensor into the cone-and-plate drive shaft for real-time feedback control, maintaining stable shear up to 50 dyne/cm². Biological validation reveals that stable shear (12 dyne/cm²) acts as a tumor suppressor in bladder cancer cells, inducing G1 arrest via p21/p27 upregulation and triggering autophagy.

[5] K. Alvarado-Estrada et al., “Circulatory shear stress induces molecular changes and side population enrichment in primary tumor-derived lung cancer cells with higher metastatic potential,” Sci. Rep., vol. 11, p. 2800, Feb. 2021, doi: 10.1038/s41598-021-82634-1.

Using a closed-loop peristaltic system, this study subjects lung cancer cells to 72 hours of circulation. The authors find that shear stress enriches for a side population of stem-like cells (4-fold increase) and drives a hybrid EMT phenotype characterized by high TWIST1/SNAI2 and CD44 expression, directly linking mechanical forces to metastatic evolution.

[6] J. Abello, S. Raghavan, Y. Y. Yien, and A. N. Stratman, “Peristaltic pumps adapted for laminar flow experiments enhance in vitro modeling of vascular cell behavior,” J. Biol. Chem., vol. 298, no. 10, p. 102404, Oct. 2022, doi: 10.1016/j.jbc.2022.102404.

This technical report compares commercial pulse dampeners with custom-built solutions. The authors demonstrate that their custom inlet/outlet dampeners achieve a six-fold suppression of pulsatile artifacts (compared to 1.4-fold for commercial units), enabling the use of standard peristaltic pumps for true laminar flow studies.

[7] S. Schneider et al., “Peristaltic on-chip pump for tunable media circulation and whole blood perfusion in PDMS-free organ-on-chip and Organ-Disc systems,” *Lab. Chip*, vol. 21, no. 20, pp. 3963–3978, 2021, doi: 10.1039/D1LC00494H.

Schneider et al. introduce the Organ-Disc, which features an integrated on-chip pumping mechanism driven by the compression of a flexible layer. This design eliminates external tubing and dead volume, allowing for the robust, self-contained perfusion of whole blood and culture media in a multiplexed disc format.

[8] K. Cenhrang et al., “3D Printed Transwell Microfluidic Devices for Epithelial Cell Culture with Shear Stress,” *ACS Meas. Sci. Au*, vol. 5, no. 4, pp. 547–558, Aug. 2025, doi: 10.1021/acsmeasuresciau.5c00045.

This paper describes a fully custom 3D-printed bioreactor featuring an electrospun collagen scaffold and a "screw-top" reservoir mechanism. The device allows for the transition from static seeding to dynamic flow (0.015-0.5 mL/min) and is compatible

with advanced imaging techniques like P-SICM, offering high-resolution morphological analysis of barrier cells under flow.

[9] I. C. Harding et al., “Developing a transwell millifluidic device for studying blood–brain barrier endothelium,” *Lab. Chip*, vol. 22, no. 23, pp. 4603–4620, Nov. 2022, doi: 10.1039/d2lc00657j.

Harding et al. present a millifluidic retrofit for standard Transwell inserts. Using this system, they demonstrate that while shear stress tightens the barrier of endothelial monocultures, it has little effect on co-cultures with pericytes and astrocytes, suggesting that multicellular complexity buffers the endothelium against mechanical variations.

[10] D. Chavarria et al., “Modular cone-and-plate device for mechanofluidic assays in Transwell inserts,” *Front. Bioeng. Biotechnol.*, vol. 13, Jan. 2025, doi: 10.3389/fbioe.2025.1494553.

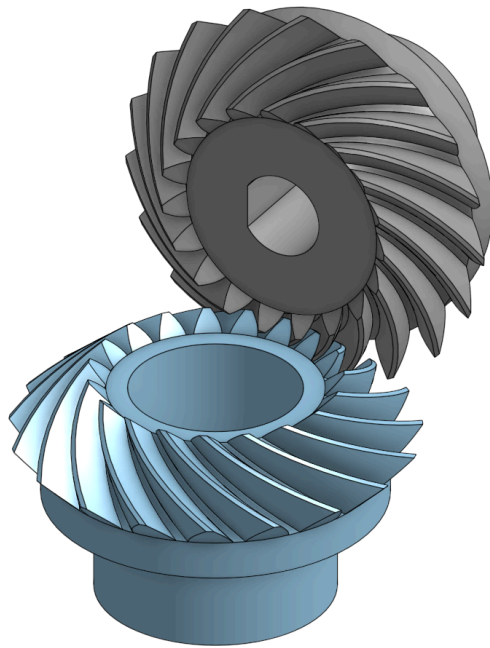
This study introduces the MoCAP device, an open-source, 3D-printed lid that adds cone-and-plate functionality to Transwell inserts. The device features interchangeable gears (1:1, 2:1) to generate multiple shear conditions (0.6-20 dyn/cm²) simultaneously. It bridges the gap between high-throughput shear assays and barrier permeability testing.

[11] K. A. S. Boster et al., “Artificial intelligence velocimetry reveals in vivo flow rates, pressure gradients, and shear stresses in murine perivascular flows,” *Proc. Natl. Acad. Sci.*, vol. 120, no. 14, p. e2217744120, Apr. 2023, doi: 10.1073/pnas.2217744120.

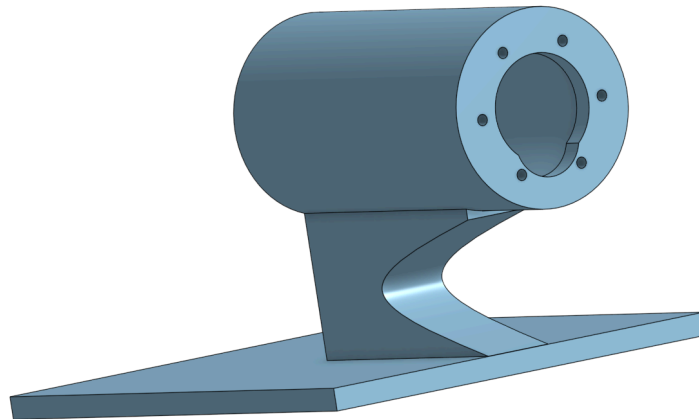
Boster et al. use Physics-Informed Neural Networks to quantify flow in the murine perivascular space. They report ultra-low mean flow speeds (16.33 $\mu\text{m/s}$) and shear stresses (0.015-0.050 dyn/cm^2), providing a critical in vivo benchmark that challenges the high shear rates used in current in vitro CNS models.

Appendix

Appendix A — Side-entry bevel gear drive CAD

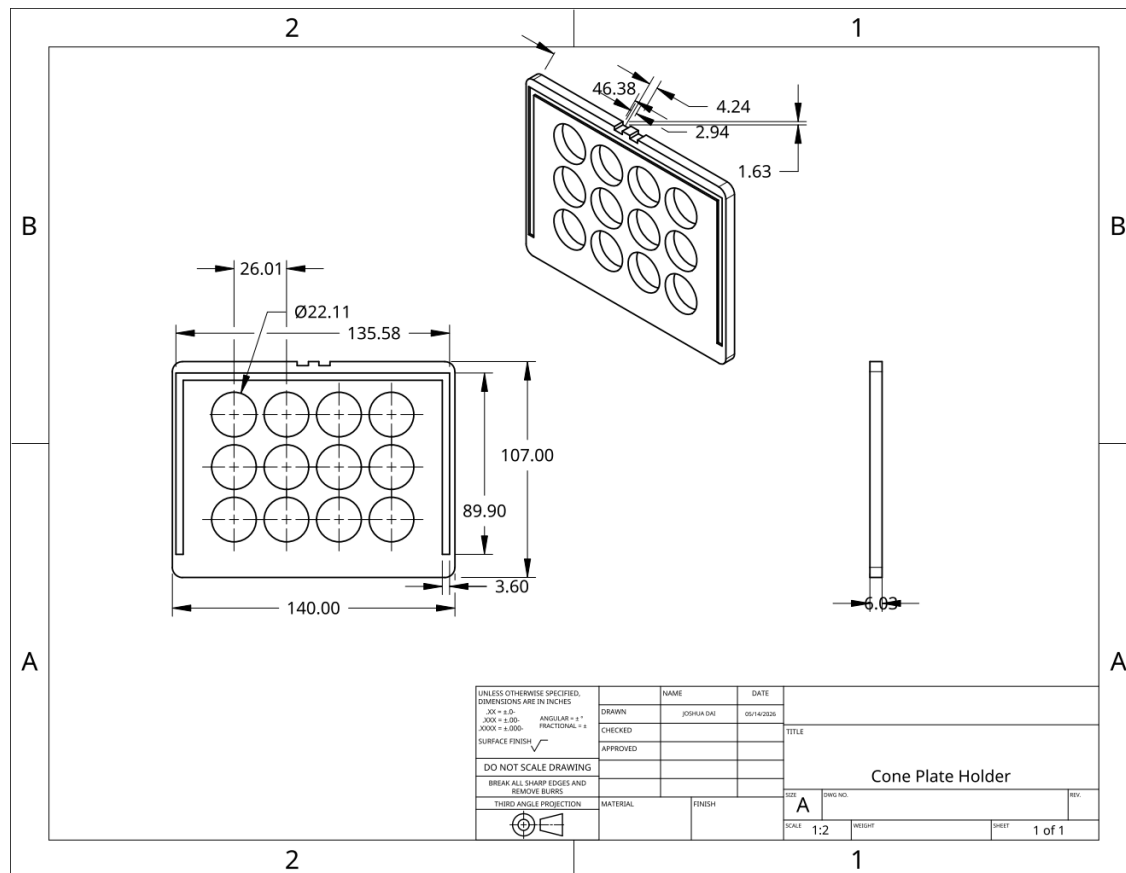


Appendix A.1: Bevel Pair Gear Drive Mechanism

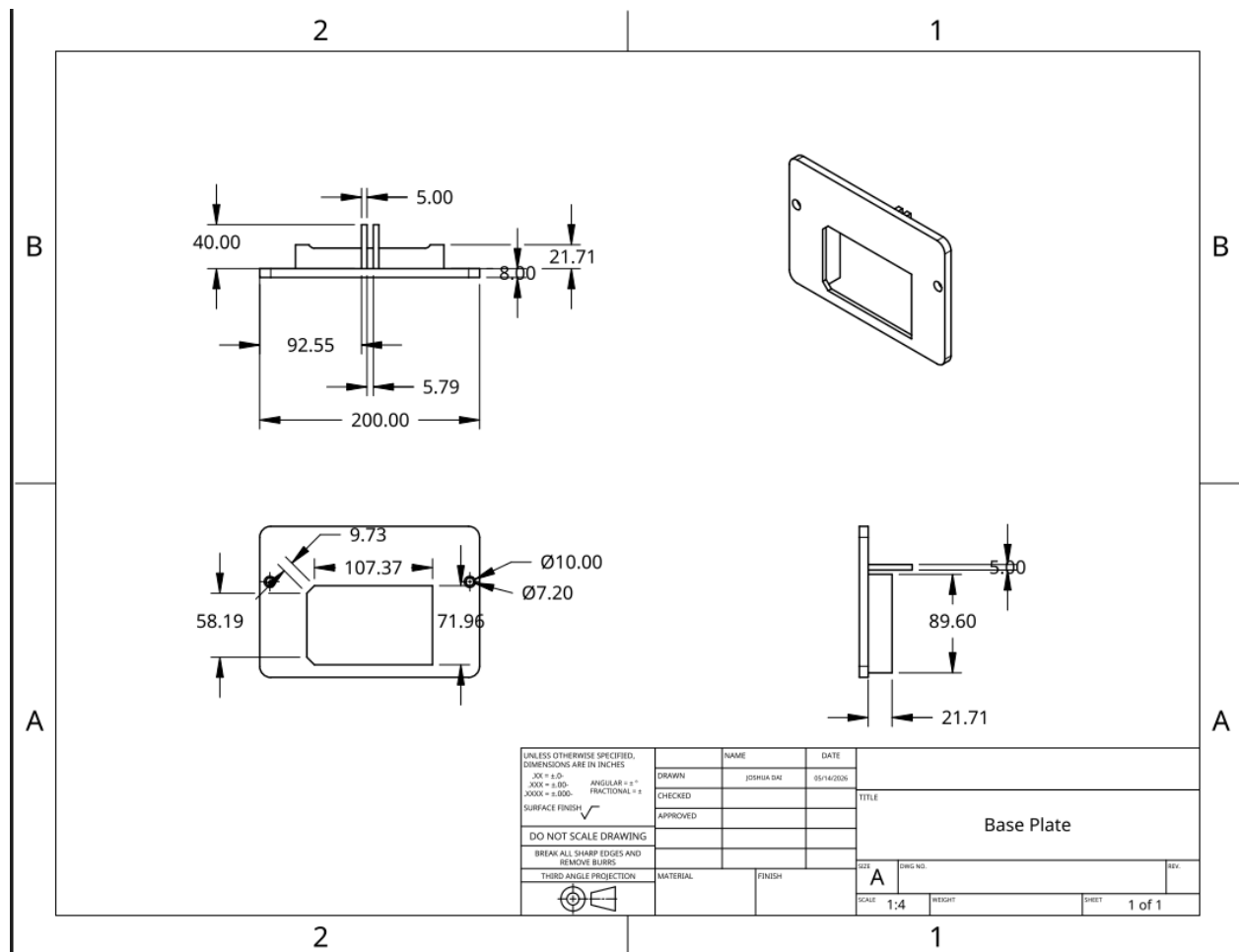


A.2: Side Holder Mount

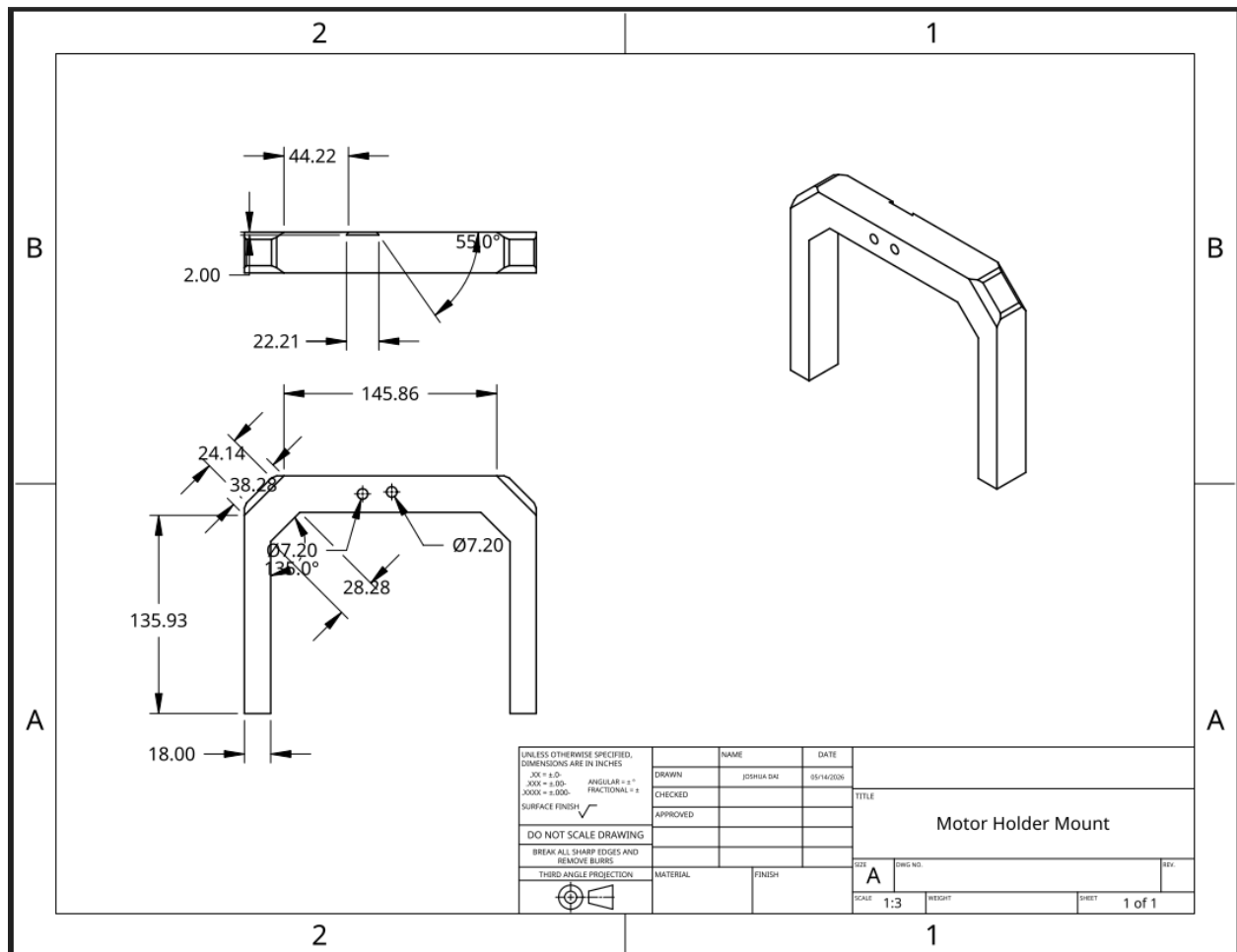
Appendix B — Final integrated assembly CAD views



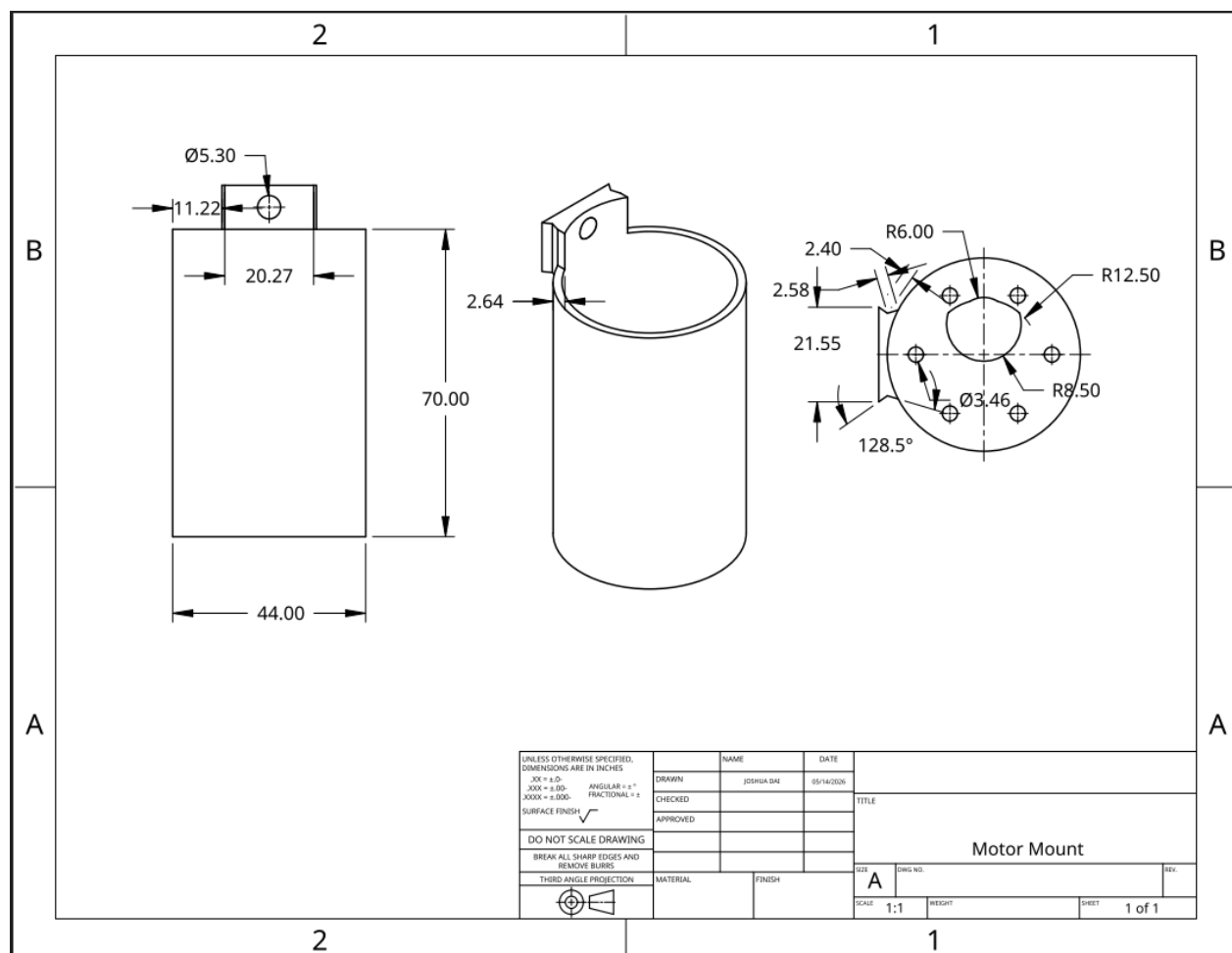
Appendix B.1: Cone Plate Holder Drawing



Appendix B.2: Base Plate Drawing



Appendix B.3: Motor Holder Mount Drawing



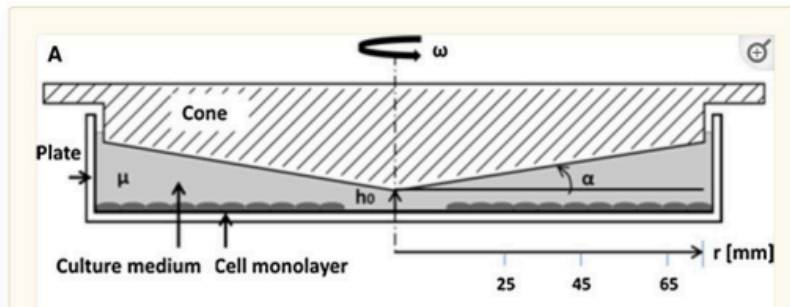
Appendix B.4: Motor Mount Drawing

Appendix C — Shear stress calculation sheet

This sheet will calculate the gear ratio in a well plate needed to produce a given shear stress with known geometry and rotational speed.

$$\omega = \frac{\tau}{\mu} \cdot \frac{h_0 + r \cdot \alpha}{r}$$

2



Slow Gear

$$\tau := 0.01 \frac{\text{dyne}}{\text{cm}^2} = 0.001 \text{ Pa}$$

$$\mu := 0.00354 \text{ Pa} \cdot \text{s}$$

$$\alpha := 0.5 \text{ deg}$$

$$h_0 := 100 \text{ } \mu\text{m} = 0.01 \text{ cm}$$

$$r := \frac{0.86969 \text{ in}}{2} = 0.435 \text{ in}$$

$$\omega := \frac{\tau}{\mu} \cdot \frac{(h_0 + r \cdot \alpha)}{r} = 0.048 \frac{\text{rev}}{\text{min}}$$

$$GR := \frac{1}{\omega} = 20.849 \frac{1}{\text{rpm}}$$

Suggested gearing will have a GR of 20.

$$GR_{actual} := 20$$

$$\omega_{actual} := \frac{1 \text{ rpm}}{GR_{actual}} = 0.05 \text{ rpm}$$

$$\tau_{actual} := \left(\frac{1}{\omega_{actual} \cdot \mu} \cdot \frac{(h_0 + (r \cdot \alpha))}{r} \right)^{-1} = 0.01 \frac{\text{dyne}}{\text{cm}^2} \quad \tau = 0.01 \frac{\text{dyne}}{\text{cm}^2}$$

$$\tau_{error} := \frac{\tau_{actual} - \tau}{\tau_{actual}} = 0.041 \quad 4.1\% \text{ error in shear.}$$

Drive gear has 24 teeth and is 1 inch in diameter.

$$N_P := 24 \quad \omega_P := 1 \text{ rpm}$$

$$N_{G4} := 48$$

$$\omega_{G4} := \frac{N_P}{N_{G4}} \cdot \omega_P = 0.5 \text{ rpm}$$

$$\omega_{G3} := \omega_{G4} = 0.5 \text{ rpm}$$

$$N_{G3} := 12$$

$$N_{G2} := 60$$

$$\omega_{G2} := \frac{N_{G3}}{N_{G2}} \cdot \omega_{G3} = 0.1 \text{ rpm}$$

$$\omega_{G1} := \omega_{G2} = 0.1 \text{ rpm}$$

$$N_{G1} := 12$$

$$N_{G0} := 24$$

$$\omega_{G0} := \frac{N_{G1}}{N_{G0}} \cdot \omega_{G1} = 0.05 \text{ rpm}$$

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Lid

Technical drawing of the Lid. The top view shows a rectangular plate with a 3x4 grid of circular holes. Dimensions include: top edge segments of 24.65 and 26.1x3; left edge segments of 16.52 and 26.1x2; overall width of 127.61±0.2; and a 45-degree chamfer on the bottom right corner. The side view shows a thickness of 8.94 and a total height of 9.94±0.15.

Plate

Technical drawing of the Plate. The top view shows a rectangular plate with a 3x4 grid of circular holes. Dimensions include: top edge segments of 25.11, 26.1, 26.1, and 26.1; left edge segments of 16.93 and 26.1; overall width of 128.53±0.2; and a 45-degree chamfer on the bottom right corner. The side view shows a total height of 20.28±0.15, with segments of 11.75 and 86.07±0.2. It also shows hole diameters of $\phi 22.21$ and $\phi 22.99$, and a central slot width of 17.27.

Features:

Appendix D.1: GenClone Specification Sheet